Analysis of Periodic Orbits in the Circular Restricted Three Body Problem

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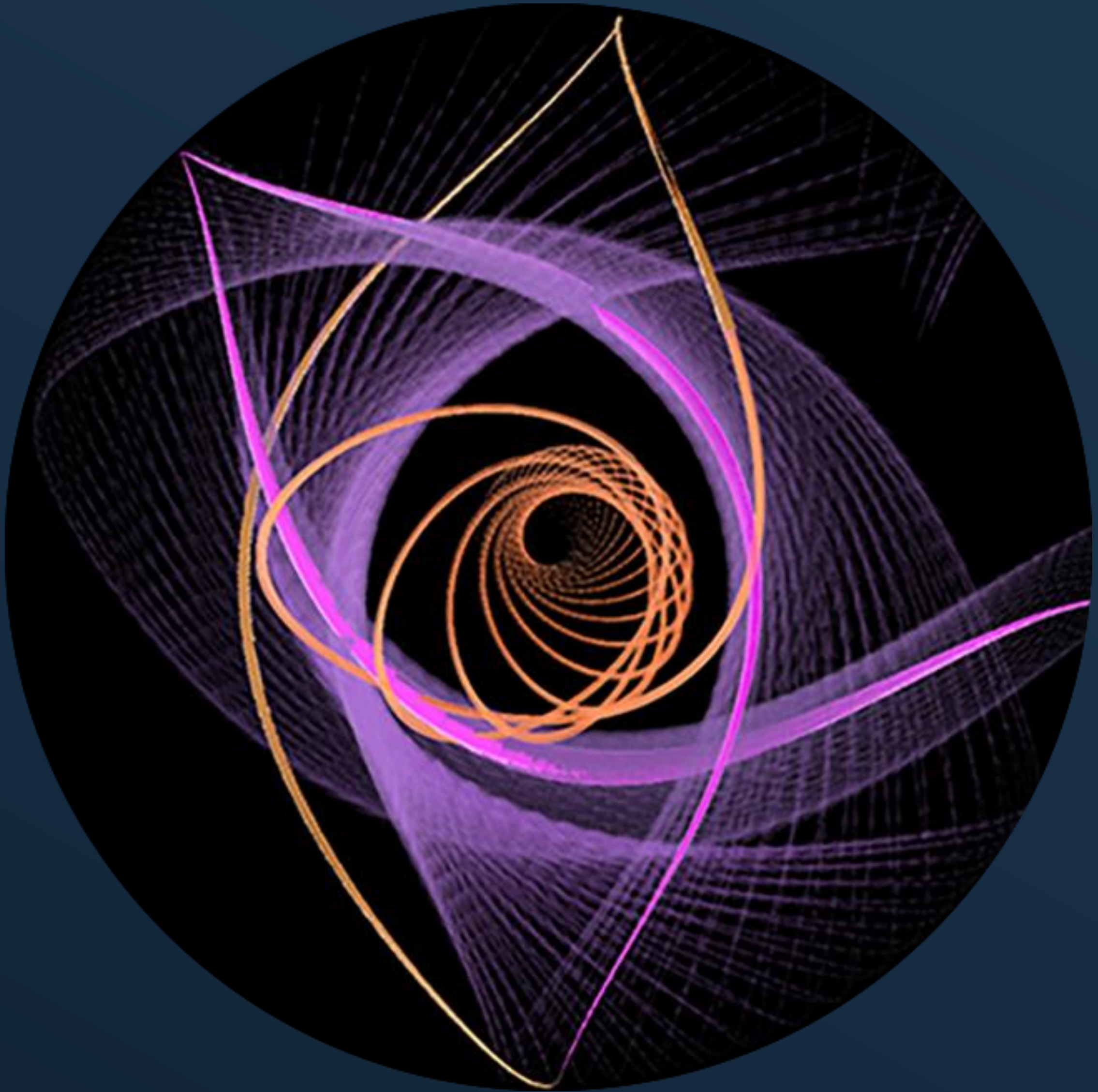
MAE 541 Project Proposal



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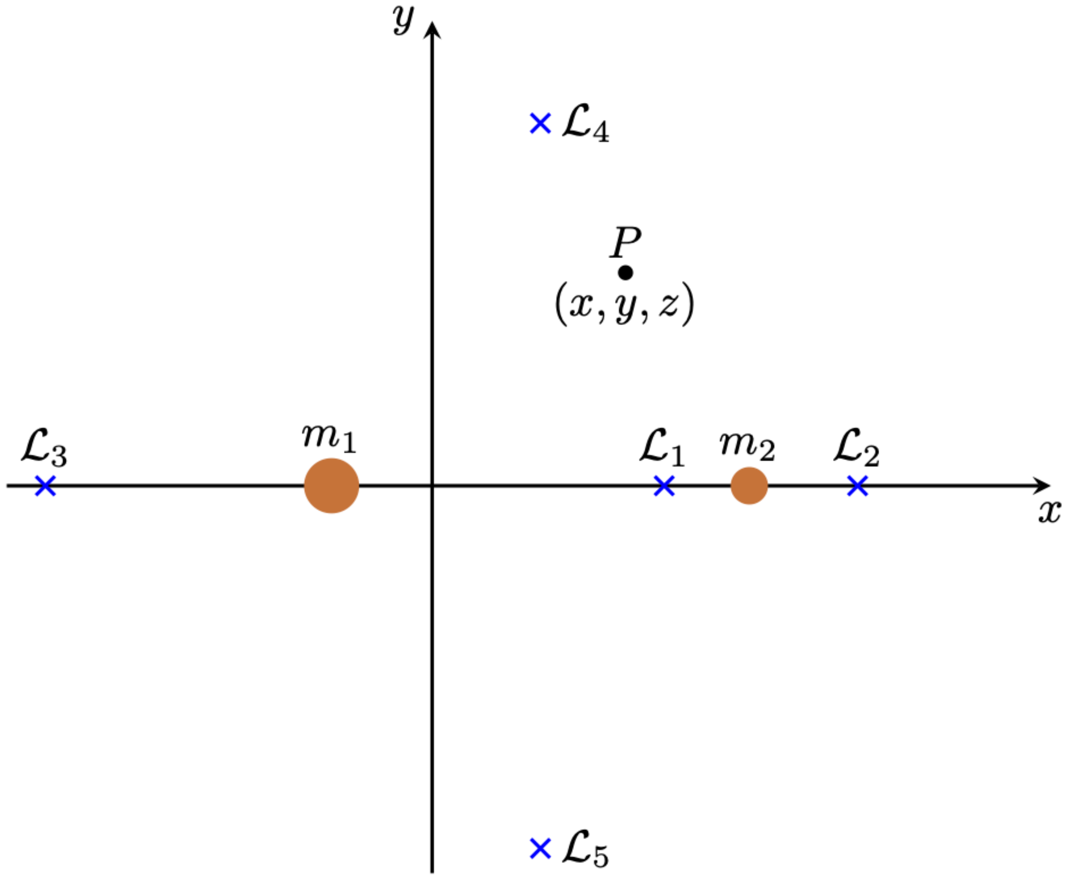
Circular Restricted 3 Body Problem (CR3BP)



- **Dynamics:**

$$\ddot{\boldsymbol{r}} = \boldsymbol{g}(\boldsymbol{r}, \boldsymbol{v}) = \begin{pmatrix} 2v_y + x - (1-\mu)\frac{x+\mu}{\rho_1^3} - \mu\frac{x-1+\mu}{\rho_2^3} \\ -2v_x + y - (1-\mu)\frac{y}{\rho_1^3} - \mu\frac{y}{\rho_2^3} \\ -(1-\mu)\frac{z}{\rho_1^3} - \mu\frac{z}{\rho_2^3} \end{pmatrix}$$

$$\rho_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}$$
$$\rho_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}$$

- **Mass parameter:** $\mu = \dfrac{m_2}{m_1 + m_2}$
- **Natural units** for mass, time and distance

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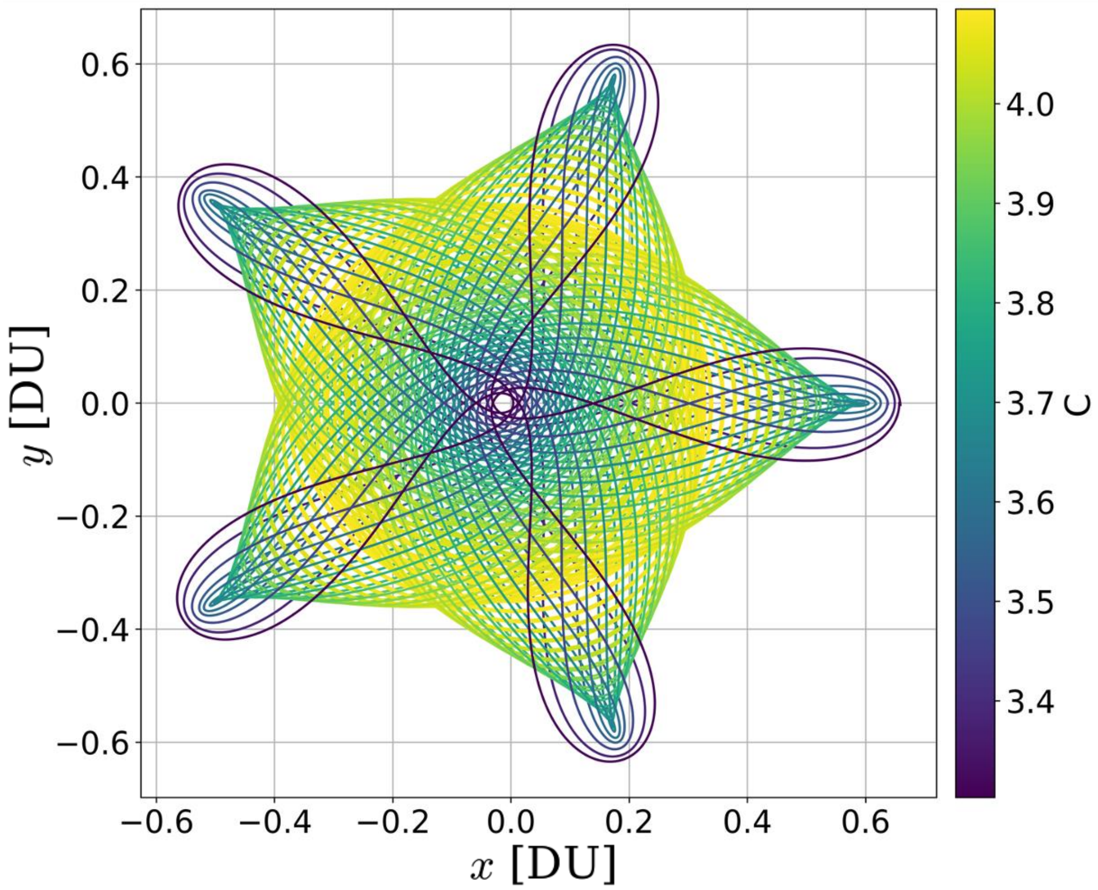
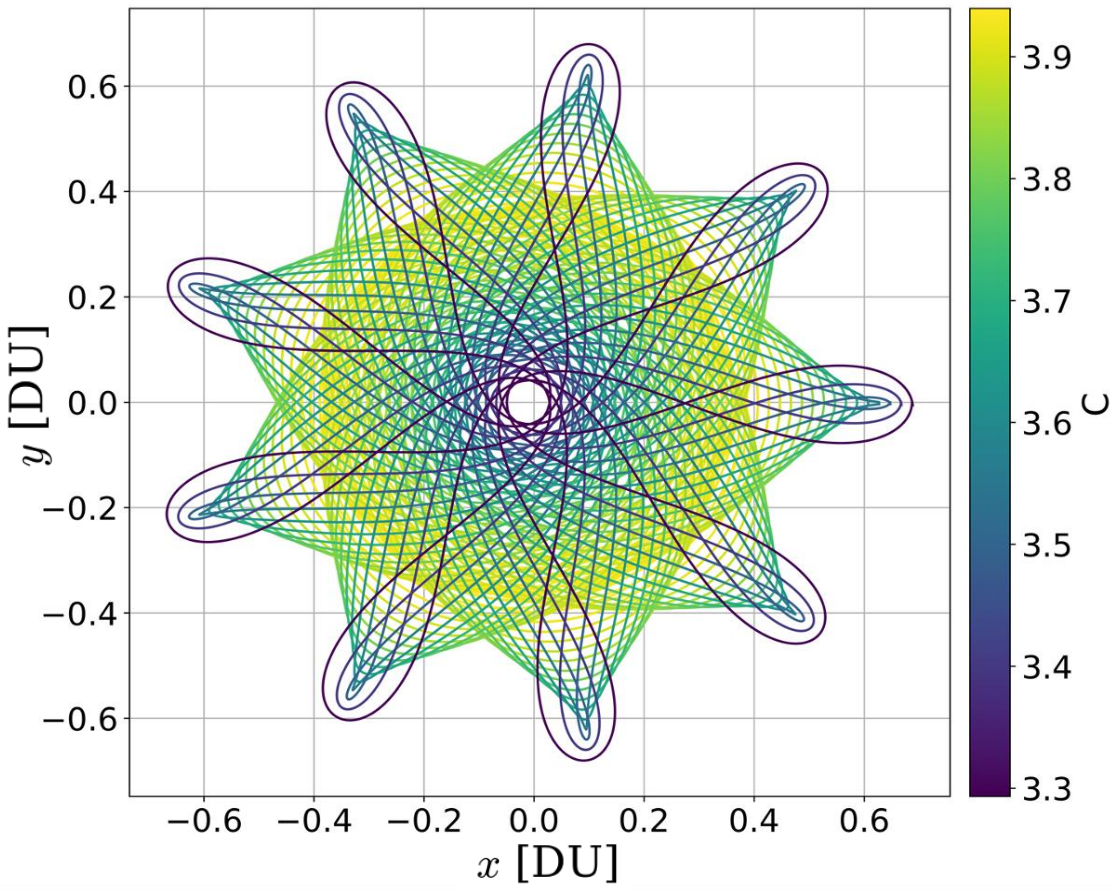
Resonant Orbits in the CR3BP

Prograde 5:1 Family



Prograde 9:2 Family



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Proposed Work

- **Derive equilibrium points of the system (Claudio):** Derive the locations of the five Lagrange points in the CR3BP and analyze their stability as a function of the mass parameter.
- **Compute periodic orbits using differential correction (Jannik):** Compute families of planar periodic orbits, including resonant and libration-point orbits, using differential correction with informed initial guesses.
- **Analyze periodic orbits on Poincaré map at** $y=0$ **(Jack):** Use a Poincaré map at $y=0$ to represent periodic orbits as fixed points and visualize their different orbital families in phase space.
- **Fixed energy Poincaré maps (Jack):** Construct fixed-energy Poincaré maps to study how the phase-space structure and degree of chaotic behavior change across energy levels.
- **Analyse bifurcations of periodic orbits (Claudio):** Investigate how families of periodic orbits change and branch through bifurcations as system parameters or energy vary.
- **Normal form around periodic orbits (Jannik):** Apply normal form transformations near a periodic orbit to simplify the local dynamics and enable more efficient propagation.

Thank you for your attention!

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